Lab 7: Fuzzing for Input Validation Bugs

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Application: shopping_cart.py (cart_total, promo_percent, delivery_zone) Tool: Hypothesis + pytest

Step 2: Original (buggy) application

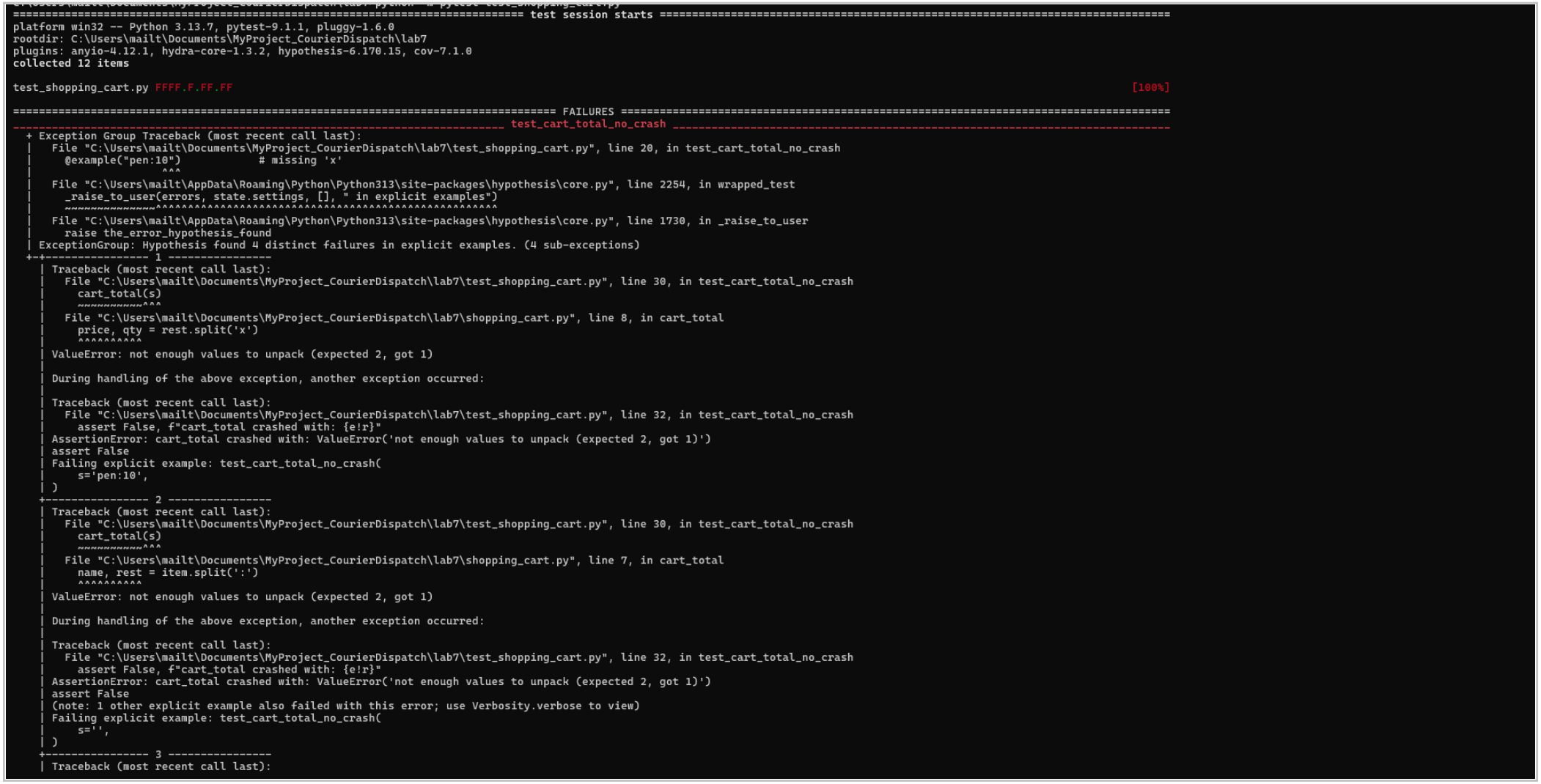
python shopping_cart.py with the cart input 'pen:10' (missing 'x') crashes with an unhandled ValueError.

```
C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7>python shopping_cart.py
==== Running cart_total ====
Enter cart as name:pricexqty CSV (e.g. pen:10x2,book:250x1): pen:10
Traceback (most recent call last):
  File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 35, in <module>
    main()
    ~~~~^^
  File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 21, in main
    result = cart_total(user_input)
  File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 8, in cart_total
    price, qty = rest.split('x')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)

C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7>
```

Step 4: Fuzz tests failing on the original code (part 1 of 7)

python -m pytest test_shopping_cart.py | Hypothesis finds crashes in cart_total (missing 'x', empty cart, non-ASCII digit, None).



Step 4: Fuzz tests failing on the original code (part 2 of 7)

python -m pytest test_shopping_cart.py | Continued: cart_total crash on '1²' and None; promo_percent crashes start.

```
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 30, in test_cart_total_no_crash
  |     cart_total(s)
  |     ~~~~~~~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 9, in cart_total
  |     total += int(price) * int(qty)
  |              ~~~^^^^^^^
  | ValueError: invalid literal for int() with base 10: '1²'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 32, in test_cart_total_no_crash
  |     assert False, f"cart_total crashed with: {e!r}"
  | AssertionError: cart_total crashed with: ValueError("invalid literal for int() with base 10: '1²'")
  | assert False
  | (note: 1 other explicit example also failed with this error; use Verbosity.verbose to view)
  | Failing explicit example: test_cart_total_no_crash(
  |     s='a:1²x1',
  | )
  +---------------- 4 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 30, in test_cart_total_no_crash
  |     cart_total(s)
  |     ~~~~~~~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 6, in cart_total
  |     for item in cart.split(','):
  |                 ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 32, in test_cart_total_no_crash
  |     assert False, f"cart_total crashed with: {e!r}"
  | AssertionError: cart_total crashed with: AttributeError("'NoneType' object has no attribute 'split'")
  | assert False
  | Failing explicit example: test_cart_total_no_crash(
  |     s=None,
  | )
  +-----------------------------------
_____________________________________________________________________ test_promo_percent_no_crash _____________________________________________________________________
 + Exception Group Traceback (most recent call last):
 |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 36, in test_promo_percent_no_crash
 |     @example("")
 |
 |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2254, in wrapped_test
 |     _raise_to_user(errors, state.settings, [], " in explicit examples")
 |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
 |     raise the_error_hypothesis_found
 | ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
 +-+---------------- 1 ----------------
   | Traceback (most recent call last):
   |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 46, in test_promo_percent_no_crash
   |     promo_percent(s)
   |     ~~~~~~~~~~~~~^^^
   |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 13, in promo_percent
   |     return int(code.upper().split('OFF')[1])
   |                ~~~~~~~~~~~~~~~~~~~~~~~~~^^^
   | IndexError: list index out of range
   |
   | During handling of the above exception, another exception occurred:
   |
```

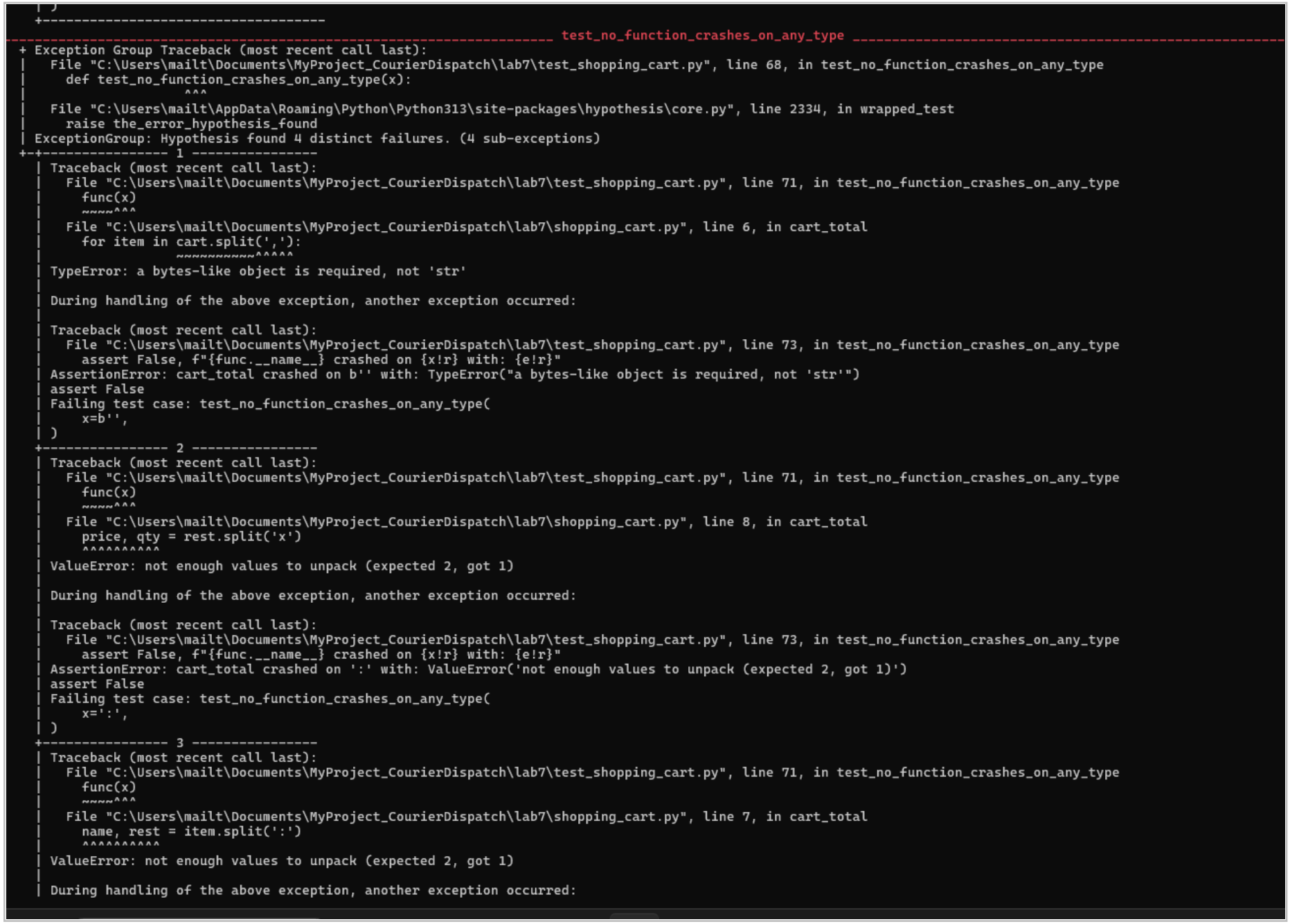
Step 4: Fuzz tests failing on the original code (part 3 of 7)

python -m pytest test_shopping_cart.py | Continued: promo_percent (empty code, 'OFF' with no number, None) and delivery_zone (empty pincode).

```
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 46, in test_promo_percent_no_crash
  |     promo_percent(s)
  |     ~~~~~~~~~~~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 13, in promo_percent
  |     return int(code.upper().split('OFF')[1])
  | ValueError: invalid literal for int() with base 10: ''
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 48, in test_promo_percent_no_crash
  |     assert False, f"promo_percent crashed with: {e!r}"
  | AssertionError: promo_percent crashed with: ValueError("invalid literal for int() with base 10: ''")
  | assert False
  | (note: 1 other explicit example also failed with this error; use Verbosity.verbose to view)
  | Failing explicit example: test_promo_percent_no_crash(
  |     s='OFF',
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 46, in test_promo_percent_no_crash
  |     promo_percent(s)
  |     ~~~~~~~~~~~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 13, in promo_percent
  |     return int(code.upper().split('OFF')[1])
  |                ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'upper'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 48, in test_promo_percent_no_crash
  |     assert False, f"promo_percent crashed with: {e!r}"
  | AssertionError: promo_percent crashed with: AttributeError("'NoneType' object has no attribute 'upper'")
  | assert False
  | Failing explicit example: test_promo_percent_no_crash(
  |     s=None,
  | )
  +------------------------------------
_____________________________________________________________________ test_delivery_zone_no_crash _____________________________________________________________________
 + Exception Group Traceback (most recent call last):
 |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 52, in test_delivery_zone_no_crash
 |     @example("")                # empty pincode
 |                  ^^^
 |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2254, in wrapped_test
 |     _raise_to_user(errors, state.settings, [], " in explicit examples")
 |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
 |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 1730, in _raise_to_user
 |     raise the_error_hypothesis_found
 | ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
 +-+---------------- 1 ----------------
   | Traceback (most recent call last):
   |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 61, in test_delivery_zone_no_crash
   |     delivery_zone(s)
   |     ~~~~~~~~~~~~~~^^^
   |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 16, in delivery_zone
   |     return ZONES[pincode[0]]
   |                  ~~~~~~~^^^
   | IndexError: string index out of range
   |
   | During handling of the above exception, another exception occurred:
```

Step 4: Fuzz tests failing on the original code (part 4 of 7)

python -m pytest test_shopping_cart.py | Continued: the any-type test crashes cart_total on bytes and on ':' input.



```
  | )
  +-----------------------------------
_____ test_no_function_crashes_on_any_type _____
+ Exception Group Traceback (most recent call last):
|   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 68, in test_no_function_crashes_on_any_type
|     def test_no_function_crashes_on_any_type(x):
|                    ^^^
|   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 71, in test_no_function_crashes_on_any_type
  |     func(x)
  |     ~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 6, in cart_total
  |     for item in cart.split(','):
  |                 ~~~~~~~~~~~^^^^^
  | TypeError: a bytes-like object is required, not 'str'
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 73, in test_no_function_crashes_on_any_type
  |     assert False, f"{func.__name__} crashed on {x!r} with: {e!r}"
  | AssertionError: cart_total crashed on b'' with: TypeError("a bytes-like object is required, not 'str'")
  | assert False
  | Failing test case: test_no_function_crashes_on_any_type(
  |     x=b'',
  | )
  +---------------- 2 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 71, in test_no_function_crashes_on_any_type
  |     func(x)
  |     ~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 8, in cart_total
  |     price, qty = rest.split('x')
  |     ^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  |
  | During handling of the above exception, another exception occurred:
  |
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 73, in test_no_function_crashes_on_any_type
  |     assert False, f"{func.__name__} crashed on {x!r} with: {e!r}"
  | AssertionError: cart_total crashed on ':' with: ValueError('not enough values to unpack (expected 2, got 1)')
  | assert False
  | Failing test case: test_no_function_crashes_on_any_type(
  |     x=':',
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 71, in test_no_function_crashes_on_any_type
  |     func(x)
  |     ~~~~^^^
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 7, in cart_total
  |     name, rest = item.split(':')
  |     ^^^^^^^^^^
  | ValueError: not enough values to unpack (expected 2, got 1)
  |
  | During handling of the above exception, another exception occurred:
```

Step 4: Fuzz tests failing on the original code (part 5 of 7)

python -m pytest test_shopping_cart.py | Continued: cart_total can raise instead of returning a valid int or None.

```
    +-----------------------------------
_____________________________________________________ test_cart_total_returns_non_negative_int_or_none _____________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 88, in test_cart_total_returns_non_negative_int_or_none
  |     def test_cart_total_returns_non_negative_int_or_none(s):
  |                ^^^
  |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 89, in test_cart_total_returns_non_negative_int_or_none
    |     result = cart_total(s)
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 8, in cart_total
    |     price, qty = rest.split('x')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_cart_total_returns_non_negative_int_or_none(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 89, in test_cart_total_returns_non_negative_int_or_none
    |     result = cart_total(s)
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 7, in cart_total
    |     name, rest = item.split(':')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_cart_total_returns_non_negative_int_or_none(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 89, in test_cart_total_returns_non_negative_int_or_none
    |     result = cart_total(s)
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 6, in cart_total
    |     for item in cart.split(','):
    |                 ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_cart_total_returns_non_negative_int_or_none(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
_____________________________________________________ test_promo_percent_rejects_over_100 _____________________________________________________

    @given(st.integers(101, 9999))
>   def test_promo_percent_rejects_over_100(n):
                ^^^

test_shopping_cart.py:99:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

n = 101

    @given(st.integers(101, 9999))
    def test_promo_percent_rejects_over_100(n):
>       assert promo_percent(f"OFF{n}") is None
E       AssertionError: assert 101 is None
E        +  where 101 = promo_percent('OFF101')
E       Failing test case: test_promo_percent_rejects_over_100(
E           n=101,  # or any other generated value
E       )

test_shopping_cart.py:100: AssertionError
```

Step 4: Fuzz tests failing on the original code (part 6 of 7)

python -m pytest test_shopping_cart.py | Continued: promo_percent accepts 'OFF101' (over 100%); delivery_zone raises KeyError for pincode starting with '0'.

```
test_shopping_cart.py:100: AssertionError
_____________________________________________________________ test_promo_percent_result_is_0_to_100_or_none _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 104, in test_promo_percent_result_is_0_to_100_or_none
  |     def test_promo_percent_result_is_0_to_100_or_none(s):
  |                    ^^^
  |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 105, in test_promo_percent_result_is_0_to_100_or_none
    |     result = promo_percent(s)
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 13, in promo_percent
    |     return int(code.upper().split('OFF')[1])
    |                ~~~~~~~~~~~~~~~~~~~~~~~~~~^^^
    | IndexError: list index out of range
    | Failing test case: test_promo_percent_result_is_0_to_100_or_none(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 105, in test_promo_percent_result_is_0_to_100_or_none
    |     result = promo_percent(s)
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 13, in promo_percent
    |     return int(code.upper().split('OFF')[1])
    |                ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'upper'
    | Failing test case: test_promo_percent_result_is_0_to_100_or_none(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________________________ test_delivery_zone_rejects_unserved_first_digit _____________________________________________________

    @given(st.sampled_from("09"), st.integers(0, 99999))
>   def test_delivery_zone_rejects_unserved_first_digit(first, rest):
                ^^^

test_shopping_cart.py:116:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
test_shopping_cart.py:117: in test_delivery_zone_rejects_unserved_first_digit
    assert delivery_zone(first + f"{rest:05d}") is None
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

pincode = '000000'

    def delivery_zone(pincode):
>       return ZONES[pincode[0]]
               ^^^^^^^^^^^^^^^^^
E       KeyError: '0'
E       Failing test case: test_delivery_zone_rejects_unserved_first_digit(
E           # The test always failed when commented parts were varied together.
E           first='0',  # or any other generated value
E           rest=0,  # or any other generated value
E       )

shopping_cart.py:16: KeyError
```

Step 4: Fuzz tests failing on the original code (part 7 of 7)

python -m pytest test_shopping_cart.py | End of the run: short test summary. Result: 9 failed, 3 passed against the original code.

```
shopping_cart.py:16: KeyError
_____________________________________________________________ test_delivery_zone_result_is_known_zone_or_none _____________________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 121, in test_delivery_zone_result_is_known_zone_or_none
  |     def test_delivery_zone_result_is_known_zone_or_none(s):
  |                    ^^^
  |   File "C:\Users\mailt\AppData\Roaming\Python\Python313\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 122, in test_delivery_zone_result_is_known_zone_or_none
    |     assert delivery_zone(s) in set(ZONES.values()) | {None}
    |            ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 16, in delivery_zone
    |     return ZONES[pincode[0]]
    |            ~~~~~^^^^^^^^^^^^
    | KeyError: '0'
    | Failing test case: test_delivery_zone_result_is_known_zone_or_none(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 122, in test_delivery_zone_result_is_known_zone_or_none
    |     assert delivery_zone(s) in set(ZONES.values()) | {None}
    |            ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 16, in delivery_zone
    |     return ZONES[pincode[0]]
    |                  ~~~~~~~~^^^
    | IndexError: string index out of range
    | Failing test case: test_delivery_zone_result_is_known_zone_or_none(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\test_shopping_cart.py", line 122, in test_delivery_zone_result_is_known_zone_or_none
    |     assert delivery_zone(s) in set(ZONES.values()) | {None}
    |            ~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7\shopping_cart.py", line 16, in delivery_zone
    |     return ZONES[pincode[0]]
    |                  ~~~~~~~~^^^
    | TypeError: 'NoneType' object is not subscriptable
    | Failing test case: test_delivery_zone_result_is_known_zone_or_none(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================================================================== short test summary info ============================================================================
FAILED test_shopping_cart.py::test_cart_total_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures in explicit examples. (4 sub-exceptions)
FAILED test_shopping_cart.py::test_promo_percent_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
FAILED test_shopping_cart.py::test_delivery_zone_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures in explicit examples. (3 sub-exceptions)
FAILED test_shopping_cart.py::test_no_function_crashes_on_any_type - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_shopping_cart.py::test_cart_total_returns_non_negative_int_or_none - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_shopping_cart.py::test_promo_percent_rejects_over_100 - AssertionError: assert 101 is None
FAILED test_shopping_cart.py::test_promo_percent_result_is_0_to_100_or_none - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_shopping_cart.py::test_delivery_zone_rejects_unserved_first_digit - KeyError: '0'
FAILED test_shopping_cart.py::test_delivery_zone_result_is_known_zone_or_none - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
========================================================================= 9 failed, 3 passed in 4.09s ==========================================================================
```

Step 5: Fixed application

python shopping_cart.py with 'pen:10', 'OFFabc' and '0': all three functions reject bad input without crashing.

```
C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7>python shopping_cart.py
==== Running cart_total ====
Enter cart as name:pricexqty CSV (e.g. pen:10x2,book:250x1): pen:10
Cart Total: Invalid cart

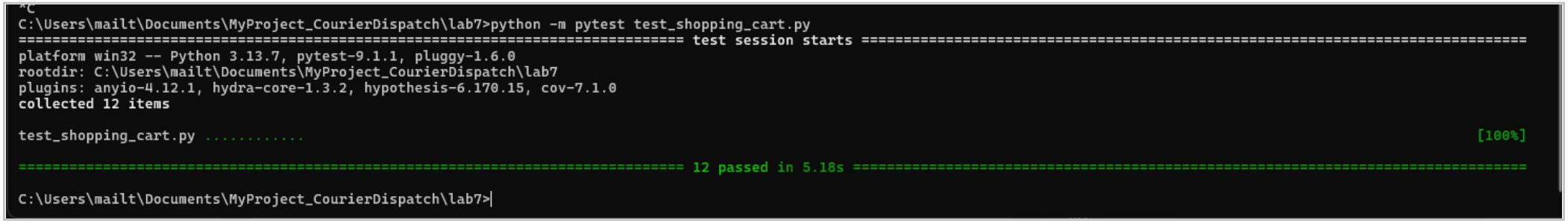
==== Running promo_percent ====
Enter a promo code (e.g. OFF20): OFFabc
Discount Percent: Invalid promo code

==== Running delivery_zone ====
Enter a 6 digit pincode (e.g. 560085): 0
Delivery Zone: Invalid or unserved pincode

C:\Users\mailt\Documents\MyProject_CourierDispatch\lab7>
```

Step 6: Fuzz tests re-run on the fixed application

python -m pytest test_shopping_cart.py: the same test file now gives 12 passed.



Summary of bugs found by fuzzing and how they were fixed

Function	Bug found (original code)	Fix
cart_total	Crashed on missing ':' or 'x', empty cart or item, non-ASCII digits, None and non-string input; accepted negative numbers.	Validate each item with a strict pattern (name:digitsxdigits); return None for any invalid cart.
promo_percent	Crashed on missing 'OFF', no number, None; accepted percentages above 100.	Strict pattern, ASCII only, 0-100 only; return None otherwise.
delivery_zone	Crashed on empty or None pincode and on first digit 0 or 9 (KeyError); accepted pincodes of any length (e.g. 5600).	Require exactly 6 ASCII digits and a served first digit; return None otherwise.
main()	Printed raw results and crashed on bad input.	Prints 'Invalid ...' messages when a function returns None.

Test file: 12 Hypothesis tests (no-crash fuzzing for each function with explicit edge-case examples, plus correctness properties).